

Time: 1hr.		Class Test-1 on Computer Graphics & Geometric Modeling, 2025	Set-1	Full Marks: 30
CO1	a.	Given the center point coordinate (8, 6) and a radius of 15 units, generate all points (in the 1st octant) by using both the Bresenham and Midpoint circle drawing algorithms. Afterward, compare the efficiency of both algorithms in terms of the number of calculations and the computational complexity involved in plotting the points.		8+2
	b.			
CO2	a.	Design a cubic Bezier curve represented by the control points p ₁ (-8, 0, 0), p ₂ (-4,5,0), p ₃ (4,-5,0), and p ₄ (8,0,0).		8+2
	b.	Define knot vector of B-Spline curves.		
CO3	a.	The lower left-hand corner and upper right-hand corner of a rectangular clipping window are at (-4, -3) and (2, 2), respectively. Using the Sutherland-Cohen algorithm, check and find the visibility portion of the line segments EF, FG, and GH for E(-5, -2), F(-2, -1), G(0, 3), and H(1, 1) in the clipping window.		10
Time: 1hr.		Class Test-1 on Computer Graphics & Geometric Modeling, 2025	Set-2	Full Marks: 30
a	a.	Given the center point coordinate (12, 8) and a radius of 18 units, generate the points that form a circle in the 2nd octant using both the Bresenham and Midpoint circle drawing algorithms. Compare the computational steps and efficiency of both algorithms in terms of their performance in generating the points.		8+2
	b.			
CO2	a.	A cubic splined curve has start and end point at (20, 0) and (7, 2) respectively. The tangent vectors at start and end points meet at (18, 9). Determine the Hermit cubic spline curve and hence find out its mid-point.		8+2
	b.	How do B-Spline curves allow local control over the shape?		
CO3	a.	The lower left-hand corner and upper right-hand corner of a rectangular clipping window are at (1, -2) and (5, 4), respectively. Using the Sutherland-Cohen algorithm, check and find the visibility portion of the line segments PQ, QR, and RS for the points P(0, 0), Q(3, 5), R(4, 3), and S(2, -1) in the clipping window.		10
Time: 1hr.		Class Test-1 on Computer Graphics & Geometric Modeling, 2025	Set-3	Full Marks: 30
CO1	a.	Consider a circle with center at (20, 15) and radius 25 units. Using both the Bresenham and Midpoint algorithms, generate the points of the circle in the 1st octant and calculate the time complexity of each algorithm based on the number of pixel calculations needed.		8+2
	b.			
CO2	a.	Find the cubic spline with two end points (1, 1) and (6, 5) and respective tangents (0, 4) and (4, 0).		6+4
	b.	Define knot vector of B-Spline curves. Suppose the knot vector of a B-spline is {0, 0, 0, 1, 2, 3, 3, 3}. Find the possible values of number of control points and degree of the polynomial.		
CO3	a.	The lower left-hand corner and upper right-hand corner of a rectangular window are at (-3, 1) and (2, 6) respectively. Using Sutherland-Cohen algorithm, check and find the visibility portion of the line segments AB, BC and CD for A(-4, 2), B(-1, 7), C(-1, 5), and D(-2, 3) in the clipping window.		10
Time: 1hr.		Class Test-2 on Computer Graphics & Geometric Modeling, 2025	Set-1	Full Marks: 30
CO4	c.	Given a CMY color (0.2, 0.4, 0.6), convert it to RGB. What does this say about color inversion? Among HSV and HLS, which is better for image editing and why?		2+2
	d.	Differentiate between Ray-casting, Ray tracing and Rediosity as rendering approach, strength, and weakness. Compare ambient light, diffuse reflection, and specular reflection in illumination models in terms of description, appearance, depending factors, strength, and weakness.		4+5
CO5	c.	Discuss how CSG operations can be applied in 3D modeling with the help of appropriate examples. Compare the usefulness of polygon meshes with CSG in representing complex solids. Also, evaluate whether wireframe models are sufficient for rendering purposes in modern graphics systems. Support your answer with valid reasoning.		6
	d.			4
CO6	b.	What strategies can be implemented to improve user acceptance of immersive technologies like VR, AR, and MR? Discuss possible non-immersive alternatives to these technologies. Why might AR be a more suitable option for creative professionals compared to VR? Support your argument with examples.		7
Time: 1hr.		Class Test-2 on Computer Graphics & Geometric Modeling, 2025	Set-2	Full Marks: 30
CO4	c.	Convert the RGB vector (200, 100, 50) to the CMY color model. What are the advantages of using the CMYK model over RGB for printing purposes? How do half-toning and dithering technique differ in terms of their visual effects and application?		2+2
	d.			3+2+4
CO5	c.	Compare the structures and applications of quadtrees and octrees in spatial data representation. What are the limitations of using wireframe models in 3D modeling and visualization?		5
	d.	Explain the key differences between Euclidean and fractal dimensions with suitable examples. How is ray-casting utilized in the implementation of CSG operations? Illustrate its role with a typical use case.		5
CO6	b.	Define morphing in the context of computer graphics. Outline the standard process involved in creating a morphing sequence. What functions does a 3D visualizer perform in multimedia applications? List and briefly describe common authoring metaphors used in multimedia systems.		7
Time: 1hr.		Class Test-2 on Computer Graphics & Geometric Modeling, 2025	Set-3	Full Marks: 30
CO4	c.	Given an RGB color vector (200, 100, 50), convert it to the corresponding YIQ color model. How does gamma correction affect the appearance of an image in terms of brightness and contrast?		2+2
	d.			2+3+4
CO5	c.	Given a Koch Curve with a segment count of 6 and a scaling factor of 1/3, calculate its total length at the 4th iteration. Based on this, determine the fractal dimension of the curve. Describe the purpose of the firing plane in ray-casting when used to implement CSG operations.		6
	d.			4
CO6	b.	In what ways does VR enhance user experience compared to traditional 3D environments? How can VR be integrated with AR to create a mixed reality experience? Discuss the challenges and limitations VR faces in widespread adoption.		7